Experiment 4

27/8/26 Setting up a Local Area network (LAN)

Aim:

To design, set up, configure LAN connecting four computers through network switch, assign static Ip addresses to all devices verify connectivity using ping command and demonstrate data sharing across the network.

Procedure:

LAN is implemented, verified using Cisco packet tracer.

Step 1: Place the devices

1. Open Cisco packet tracer
2. From device category panel, select end devices drag 4PC onto workspace
3. Select Switches, drag 1 Switch onto to workspace

Step 2: Cable the devices

4. Click Connections → Select Copper - Straight - Through cable
5. Click each PC, then click button Select an available FastEthernet port to complete each connection
6. Repeat for all four PC's. Green link lights confirm successful connection

Step 3: Assign Ip addres to PCs

7. Click a PC → Desktop tab → Ip configuration
8. Enter Static Ip address, Subnet mask as per Ip tables for each PC.

Step 4: Configure the Switch Management Ip

9. Click the Switch → CLI tab

10. Enter the following

```
Configure terminal
Interface Vlan 1
ip address 10.1.1.5.255.0.0.0
NO Shutdown
exit
```

Step 5: Verify connectivity using ping

11. Click a PC → Desktop tab → command prompt.

12. Use ping command to test connectivity to other PC's

13. Successfull "Reply from" message confirms connectivity

Step 6: Demonstrate data sharing.

14. Add a server device, connect it to switch

15. Assign an Ip address

16. Open server's services

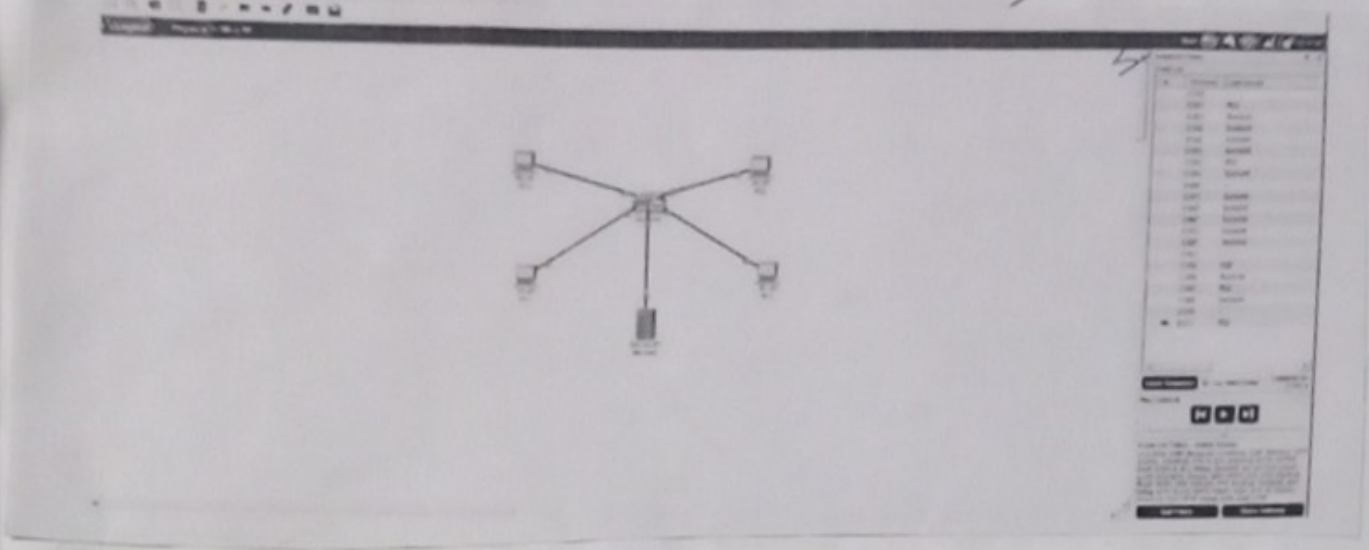
17. From another PC, open web browser, enter

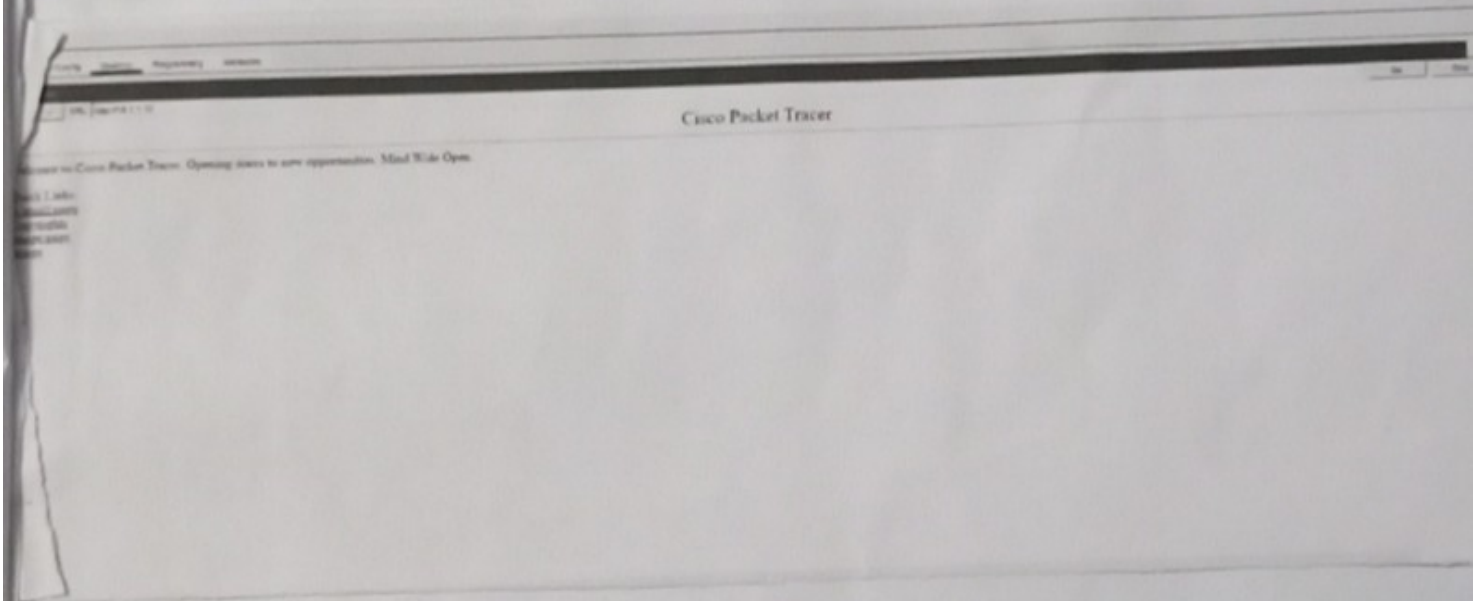
18 address to load the page, confirming successful data transfer across LAN.

Step 7: Observe Packet flow in Simulation mode.

18. Switch to Simulation mode

19. Send a ping or web request between two devices, use play / capture to step through the packet as it travels hop by hop through switch, observing MAC address based forwarding





Result:

Thus the design, setup, configure a LAN and has successfully verified Connectivity using ping command and demonstrated resource network.